Course Name: Differential Equations
Course Code: MA-121
Course Type: Institute Core

Contact Hours/Week: 3L **Course Credits:** 03

Course Objectives:

- To introduce the fundamental concepts relevant to Ordinary & Partial Differential Equations, Transform Theory.
- To able to form and solve the ordinary & partial differential equation using different analytical techniques.
- To have the idea of Laplace transform and its uses in engineering problems.

Unit Number	Course Content	Contact Hours
Unit-01	**First Order Ordinary Differential Equations:** Exact Differential Equations, Equations reducible to exact equations, Non-linear Equation of First Order, Clairaut's equation.	05L
Unit-02	**Higher Order Ordinary Differential Equations:** Higher order linear differential equations with constant co-efficient, Complimentary functions and particular integral, Method of variation of parameters, Equations reducible to linear differential equations with constant co-efficient (Cauchy-Euler and Legendre's linear differential equations), Simultaneous linear equations with constant co-efficient.	07L
Unit-03	**Ordinary Differential Equations with Laplace Transform:** Laplace transform, inverse Laplace transform, and their properties, Solution of ODE and linear simultaneous differential equations using Laplace transform.	06L
Unit-04	**First Order Partial Differential Equations:** Introduction and formulation of Partial Differential Equations (PDE), Solution of PDE, Linear PDE of First Order (Lagrange's Linear Equation), Non-linear Equation of First Order (Standard Forms), Charpit's Method.	06L
Unit-05	**Higher Order Partial Differential Equations:** Homogeneous Linear Equations with Constant Coefficients, Non-homogeneous Linear Partial Differential Equations, Non-linear partial differential equation of second order (Monge's Method).	06L
Unit-06	**Applications of Partial Differential Equations:** Classification of second order PDEs, Method of separation of variables, Solution of one dimensional wave and heat equation, Solution of two dimensional Laplace's equation.	06L

Course Outcomes:

Upon successful completion of the course, the student will be able to

CO1: Understand and analysis the theoretical & practical aspects of ODE, PDE, and Laplace Transform theory.
CO2: Identify an appropriate technique to solve the ODE, PDE.
CO3: Learning the limitations and advantages of ODE, PDE, and Laplace transform.
CO4: Apply the concepts of ODE, PDE, and Laplace transform in various engineering problems.
CO5: Demonstrate the concepts through examples and applications.

Text Books:

1. E. Kreyszig, Advanced Engineering Mathematics, John Wiley and Sons, NC, New York.
2. Paras Ram, Engineering Mathematics: Through Applications, CBS Publisher & Distributors Pvt. Ltd.
3. Henner Victor, Belozerova Tatyana, Khenner Mikhail, Ordinary and Partial Differential Equations, CRC Press.

Reference Books:

1. S.L. Ross, Differential Equations, John Wiley & Sons, New York.
2. Lawrence C.Evans, Partial Differential Equations, American Mathematical Society.
3. M. D Raisinghania, Ordinary and Partial Differential Equations, S. Chand & Company Ltd.